

LINUX ASSIGNMENT – DAY 2

Q1. Create the recursive directory structure in /opt.

```
[root@jonathan-linux-assignment02 ~]# tree /opt/linux_lab/
/opt/linux_lab/
├── applications
│   ├── backend
│   │   ├── logs
│   │   └── src
│   └── frontend
│       ├── config
│       └── src
├── backup
├── reports
│   ├── daily
│   └── monthly
└── 
```

11 directories, 0 files
[root@jonathan-linux-assignment02 ~]#

Q2. Create files with different sample content.

```
[root@jonathan-linux-assignment02 ec2-user]# ls /opt/linux_lab/
applications  backup  reports
[root@jonathan-linux-assignment02 ec2-user]# tree /opt/linux_lab/
/opt/linux_lab/
├── applications
│   ├── backend
│   │   ├── logs
│   │   │   └── backend.log
│   │   └── src
│   │       ├── file1.txt
│   │       ├── file2.txt
│   │       └── file3.txt
│   └── frontend
│       ├── config
│       │   ├── config1.conf
│       │   └── config2.conf
│       └── src
│           ├── file1.txt
│           ├── file2.txt
│           └── file3.txt
├── backup
├── reports
│   ├── daily
│   └── monthly
└── 
```

11 directories, 9 files
[root@jonathan-linux-assignment02 ec2-user]#

Q3. Copy applications to backup and update only newer files.

```
33 sudo cp -r /opt/linux_lab/applications /opt/linux_lab/backup/
34 echo "Updated frontend source" >> /opt/linux_lab/applications/frontend/src/file1.txt
35 sudo cp -ru /opt/linux_lab/applications/. /opt/linux_lab/backup/applications/
36 clear
```

Q4. Move backend/src contents to frontend/src and rename a file.

```
39 sudo mv /opt/linux_lab/applications/backend/src/file1.txt /opt/linux_lab/applications/frontend/src/
40 sudo mv /opt/linux_lab/applications/backend/src/file2.txt /opt/linux_lab/applications/frontend/src/
41 sudo mv /opt/linux_lab/applications/backend/src/file3.txt /opt/linux_lab/applications/frontend/src/
42 sudo mv /opt/linux_lab/applications/frontend/src/file1.txt /opt/linux_lab/applications/frontend/src/backend-file1.txt
```

Q5. Investigate hard links.

```
└─ reports
   └─ daily
      ├── report-hardlink1.txt
      └── report-hardlink2.txt
   └─ monthly
```

```
sudo touch /opt/linux_lab/reports/daily/report.txt
sudo ln /opt/linux_lab/reports/daily/report.txt /opt/linux_lab/reports/daily/report-hardlink1.txt
sudo ln /opt/linux_lab/reports/daily/report.txt /opt/linux_lab/reports/daily/report-hardlink2.txt
ls -li /opt/linux_lab/reports/daily/report.txt
ls -li /opt/linux_lab/reports/daily/report-hardlink1.txt
ls -li /opt/linux_lab/reports/daily/report-hardlink2.txt
ls -l /opt/linux_lab/reports/daily/report.txt
echo "Updated report" >> /opt/linux_lab/reports/daily/report.txt
cat /opt/linux_lab/reports/daily/report-hardlink1.txt
cat /opt/linux_lab/reports/daily/report-hardlink2.txt
sudo rm /opt/linux_lab/reports/daily/report.txt
cat /opt/linux_lab/reports/daily/report-hardlink1.txt
cat /opt/linux_lab/reports/daily/report-hardlink2.txt
```

Q6. Investigate symbolic links.

```
8 sudo rm -rf /opt/linux_lab/reports/monthly/test
9 sudo ln -s /opt/linux_lab/reports/monthly /opt/linux_lab/monthly-link
10 sudo touch /opt/linux_lab/monthly-link/test.txt
11 ls -l /opt/linux_lab/monthly-link/test.txt
12 readlink -f /opt/linux_lab/monthly-link/test.txt
13 sudo rm -rf /opt/linux_lab/reports/monthly
14 ls -l /opt/linux_lab/monthly-link
```

Q7. Compare hard link and soft link.

```
7 sudo touch /opt/linux_lab/reports/daily/original.txt
8 sudo ln /opt/linux_lab/reports/daily/original.txt /opt/linux_lab/reports/daily/hardlink.txt
9 sudo ln -s /opt/linux_lab/reports/daily/original.txt /opt/linux_lab/reports/daily/softlink.txt
10 ls -li /opt/linux_lab/reports/daily/original.txt
11 ls -li /opt/linux_lab/reports/daily/hardlink.txt
12 ls -li /opt/linux_lab/reports/daily/softlink.txt
13 readlink /opt/linux_lab/reports/daily/softlink.txt
14 sudo rm /opt/linux_lab/reports/daily/original.txt
15 ls -l /opt/linux_lab/reports/daily/hardlink.txt
16 ls -l /opt/linux_lab/reports/daily/softlink.txt
```

Q8. Create appadmin with a custom UID, home directory and Bash shell.

```
[root@jonathan-linux-assignment02 ~]# grep '^appadmin:' /etc/passwd
appadmin:x:2501:2501:~/home/appadmin:/bin/bash
[root@jonathan-linux-assignment02 ~]#
```

Q9. Create groups and configure user memberships.

```
[root@jonathan-linux-assignment02 ~]# id dev2
id: 'dev2': no such user
[root@jonathan-linux-assignment02 ~]# id test1
uid=2503(test1) gid=2504(test1) groups=2504(test1),1005(testers)
[root@jonathan-linux-assignment02 ~]# id ops1
uid=2504(ops1) gid=2505(ops1) groups=2505(ops1),2502(operations)
[root@jonathan-linux-assignment02 ~]#
```

Q10. Apply user restrictions to test1.

```
Locking password for user test1.
passwd: Success
test1 LK 2026-08-31 0 99999 7 -1 (Password locked.)
Last password change                : Aug 31, 2026
Password expires                    : never
Password inactive                   : never
Account expires                    : Dec 31, 2026
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
Unlocking password for user test1.
passwd: Warning: unlocked password would be empty.
passwd: Unsafe operation (use -f to force)
[root@jonathan-linux-assignment02 ~]#
```

```
9 sudo passwd -l test1
10 sudo passwd -S test1
11 sudo chage -E 2026-12-31 test1
12 sudo chage -l test1
13 sudo passwd -u test1
```

Q11. Create and configure the shared project directory.

```
[root@jonathan-linux-assignment02 ~]# ls -ld /opt/shared_project
drwxrws---. 2 root developers 6 Aug 31 18:03 /opt/shared_project
[root@jonathan-linux-assignment02 ~]#
```

Q12. Configure user-specific ACL permissions for application.conf.

```
[root@jonathan-linux-assignment02 ~]# getfacl /opt/shared_project/application.conf
getfacl: Removing leading '/' from absolute path names
# file: opt/shared_project/application.conf
# owner: root
# group: developers
user::rw-
user:dev1:rw-
user:test1:r--
user:ops1:rw-
group::r--
mask::rw-
other::---

[root@jonathan-linux-assignment02 ~]#
```

Q13. Configure default ACL permissions and verify inheritance.

```
[root@jonathan-linux-assignment02 ~]# getfacl /opt/shared_project/newfile.txt
getfacl: Removing leading '/' from absolute path names
# file: opt/shared_project/newfile.txt
# owner: root
# group: developers
user::rw-
user:dev1:rw-
user:test1:r--
user:ops1:rw-
group::rwx #effective:rw-
mask::rw-
other::---

[root@jonathan-linux-assignment02 ~]#
```

Q14. Configure ownership and permissions recursively.

```
[root@jonathan-linux-assignment02 ~]# ls -lR /opt/application/data
/opt/application/data:
total 0
-rw-rw----. 1 appadmin developers 0 Aug 31 18:06 file1.txt
-rw-rw----. 1 appadmin developers 0 Aug 31 18:06 file2.txt
-rw-rw----. 1 appadmin developers 0 Aug 31 18:06 file3.txt
[root@jonathan-linux-assignment02 ~]#
```

Q15. Troubleshoot dev1 unable to modify an appadmin-owned file.

```
[root@jonathan-linux-assignment02 ~]# ls -l /opt/application/data/file1.txt
-rw-rw----. 1 appadmin developers 0 Aug 31 18:06 /opt/application/data/file1.txt
[root@jonathan-linux-assignment02 ~]#
```